16 For all values of x

$$f(x) = \frac{x+3}{5} \quad \text{and} \quad g(x) = (x-7)^2$$

(a) Find $f^{-1}(x)$

Let $y = (x + 3)/5$

1 mark
Reverse the operations.

$$5y = x + 3$$
$$-3 \qquad -3$$

$x = 5y - 3$, so $f^{-1}(x) = 5x - 3$

1 mark
Write the inverse using x as the input.

$f^{-1}(x) =$ $5x - 3$

(2)

(b) Find gf(47)

$f(47) = (47 + 3) \div 5 = 10$

$gf(47) = g(10) = (10 - 7)^2 = 9$

1 mark
Apply f first, then g to its output.

1 mark
Square 10 – 7.

9

(2)

(Total for Question 16 is 4 marks)

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